# Strategic Analysis of NYC Citi Bike Service 2022

A Data-Driven Deep Dive into Urban Mobility Imbalances and Operational Optimization

**Case Study: Dr. Lylan Bramasole**

# The Challenge: Network Imbalance


## Recurring Service Bottlenecks

Citi Bike faces a critical operational challenge: high-demand stations frequently encounter **"dock-outs"** (empty) or **full docks** during peak hours.

These imbalances disrupt the user experience, particularly for commuters, and necessitate high-cost manual fleet rebalancing via vans and valet services.

# Integrated Data Foundation

By enriching 20 million trip records with environmental metrics, I  identify the primary drivers of demand variance.

| Data Source | Metric Categories | Volume / Granularity |
|---|---|---|
| Citi Bike Open Data | Timestamps, GPS, Station Names, Rider Type | 20,000,000+ Records (2022) |
| NOAA Weather API | Temperature (TAVG), Precipitation, Wind Speed | Daily Averages (LaGuardia Station) |
| Engineered Data | Route IDs, Duration Clusters, Peak Hour Tiers | Cleaned Python Aggregations |

# Methodology: The Analytical Pipeline

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## Data Ingestion

Concatenating 12 monthly CSV files via AWS S3.

## Preprocessing

Cleaning GPS errors & outliers; Datetime parsing.

## Integration

Merging trip data with NOAA weather API keys.

## Analysis

Spatial clustering & seasonal trend modeling.

## Deployment

Interactive dashboard built with Streamlit.

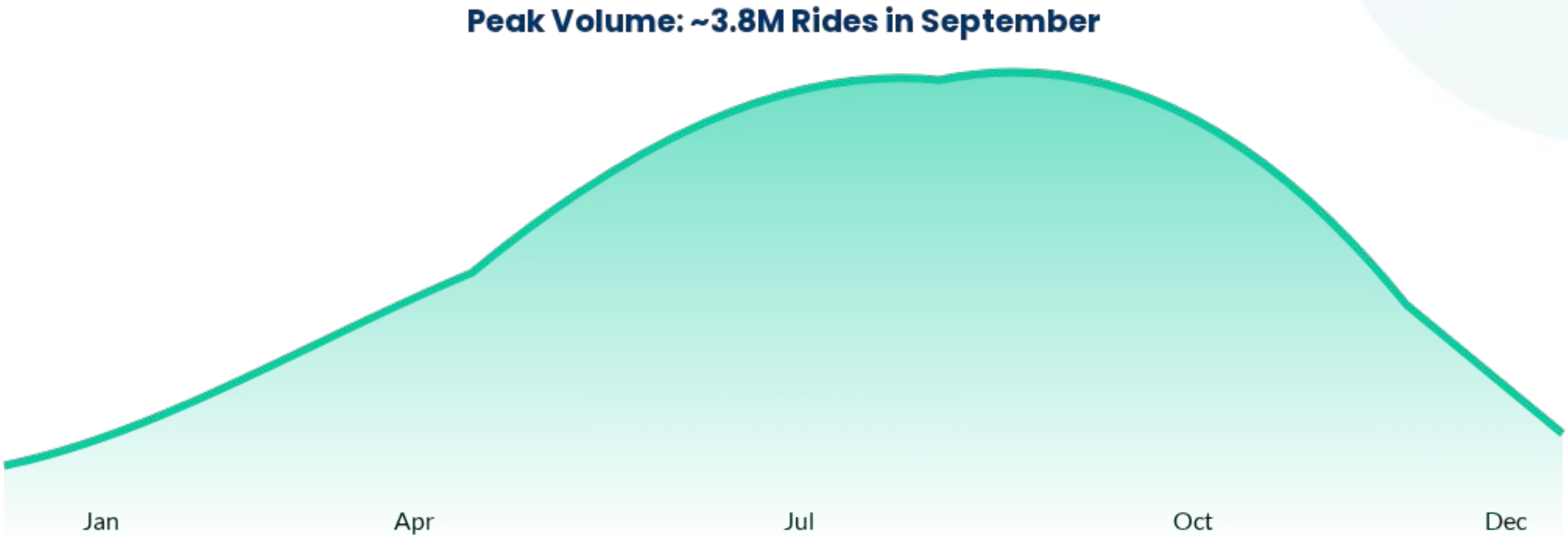
# Results & Insights

Uncovering the Spatiotemporal Dynamics of NYC Cycling

# Station Demand Concentration

Top 10 Stations by Departure Volume. Demand is heavily skewed toward transit hubs and business districts.

# Ridership Peaks with Temperature




*Daily analysis confirms a direct linear correlation between Average Temperature and trip volume.*

# Route Dynamics: Commuter vs. Leisure

## Weekday Patterns

The system acts as a **commuter backbone**. Flows are highly directional: Residential areas to Business Districts (AM) and the inverse during (PM) rush hours. Imbalance is most acute at 8-9 AM and 5-6 PM.

## Weekend Patterns

Activity shifts toward **leisure loops**. High concentration around Central Park, Hudson River Greenway, and waterfront paths in Brooklyn. Routes are often non-linear, beginning and ending at the same station.

# The Extreme Long-Tail Distribution

75%

of unique routes occur only once

## Operational Efficiency Focus

Despite millions of trips, the majority of unique point-to-point connections are rare events.

Strategy must prioritize the **high-frequency corridors** that account for the bulk of ridership. Efficiency is won or lost in the top 1% of routes.

# Geographic Gaps

## Outer Borough Opportunities

While Manhattan is saturated with stations, significant portions of **Queens, Brooklyn, and the Bronx** remain "service deserts."

Scaling dock density in these high-potential residential zones can bridge the gap between leisure riding and daily utility transit.

# Strategic Recommendations

### Optimize Rebalancing

Target manual redistribution to the top 20 high-frequency commuter corridors.

### Seasonal Maintenance

Utilize the Jan-Feb ridership dip for heavy fleet overhaul and repair cycles.

### Data-Led Expansion

Prioritize new station density in transit-dense outer borough corridors.

# Questions?

Thank you for your attention to this strategic analysis.

**Explore Interactive Dashboard:**

nyccitibike-analysis.streamlit.app

**GitHub Repository:** github.com/L-Bramasole/NYC_CitiBike

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